

The Connecting Thread Theory: Phenomenological Evaluation of Dimensional Leakage and Kinematic Divergence at the 215.111 GeV Thermodynamic Vacuum Ceiling

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Abstract

We present a comprehensive phenomenological evaluation of the 215.111 GeV Critical Vacuum Ceiling (R_{crit}) postulated by the Connecting Thread Theory (TBP) using 13 TeV pp collision datasets from CERN Open Data. To ensure instrumental consistency, a dataset of billions of events was sanitized via XRootD utilizing a Golden JSON certification and six primary Missing Transverse Energy (E_T^{miss}) noise filters to eliminate cosmic and calorimeter anomalies. A multi-channel extraction pipeline evaluated eight independent topologies simultaneously.

A multiparametric statistical evaluation revealed significant anomalous deviations localized at the 212–218 GeV range. Profile Likelihood Ratios exposed distinct structural divergences in the heavy-mass JetHT (-12.14σ) and light-mass Monophoton sectors. Welch's T-Test quantified E_T^{miss} dimensional leakage intensities: Monophoton at 19.77σ (0.87 GeV shift), MuonEG at 13.81σ , and Di-Photon at 10.17σ . These divergences scaled linearly according to $\sqrt{N}$ luminosity evolution. Crucially, to validate the fundamental nature of the signal, a rigorous robustness evaluation utilizing the complete suite of standard CMS High-Level Triggers (HLT) and Tight electronic Identification protocols was executed. Rather than attenuating the signal, the thermodynamic divergence robustly survived this stringent filtering, yielding an -18.23σ causality anomaly in the JetHT sector and a 7.28σ W-Test divergence in the Single-Electron sector. Methodological closure via null-hypothesis tests further excluded systematic extraction artifacts. The results provide mutually consistent, multi-sector phenomenological support for the TBP geometric architecture.

Keywords: Missing Transverse Energy, Kinematic Divergence, Dimensional Leakage, Critical Vacuum Ceiling, Heaviside Truncation, Connecting Thread Theory

1. Introduction

The Standard Model of particle physics relies on phenomenological parameterization within a continuous, infinitely differentiable spacetime manifold to resolve ultraviolet divergences [3]. Despite high-luminosity operations at the Large Hadron Collider (LHC), searches for heavy invariant mass resonances frequently yield null results [3]. This observational absence suggests potential structural limitations in extending continuous kinematic manifolds to localized, extreme energy densities.

The Connecting Thread Theory (TBP) proposes a geometric framework that replaces the search for novel heavy particles with the postulate of a finite thermodynamic manifold limit [1]. TBP hypothesizes the existence of a Critical Vacuum Ceiling (R_{crit}) at precisely 215.111 GeV. Under this theoretical architecture, the operational manifold is truncated by a Heaviside step function, $\Theta(R_{crit} - \mathcal{M})$, where $\mathcal{M}$ is the invariant mass of the localized quantum system. The kinematic boundary condition is governed by the distributional derivative of this constraint, yielding a Dirac delta singularity representing a thermodynamic

boundary:

$$\frac{\partial}{\partial \mathcal{M}} \Theta(R_{crit} - \mathcal{M}) = -\delta(R_{crit} - \mathcal{M}) \quad (1)$$

When a system reaches this δ -function boundary, momentum exceeding the threshold cannot be sustained within the standard four-dimensional projection and transitions into orthogonal dimensions—a mechanism defined as dimensional leakage [1]. In the macroscopic topologies of high-energy collider physics, TBP predicts that this dimensional transition will manifest phenomenologically as distinct deviations in Missing Transverse Energy (E_T^{miss}). This manuscript provides a multi-sector phenomenological evaluation of the TBP geometrical postulate by analyzing 13 TeV collision events, probing for simultaneous topological deformation and kinematic divergence at the 215.111 GeV threshold.

2. Methodology and Operational Protocols

To phenomenologically test the TBP framework, the conservation of the 6-dimensional Energy-Momentum Tensor must be evaluated. TBP dictates that momentum lost in 4D spacetime ($\partial_\mu T_{4D}^{\mu\nu}$) equates to the flux traversing the internal dimensional boundary ($-\partial_a T_{2D}^{av}$). To assess this, a multi-channel extraction pipeline was deployed across 13 TeV proton-proton collision datasets from the CERN Open Data portal [2].

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2.1. Phase I: The Purity Protocol and Noise Elimination

Data sanitization was governed by a Golden JSON certification [4], restricting the extraction to luminosity blocks where all detector subsystems operated without defect. To eliminate non-collision artifacts capable of falsifying E_T^{miss} vectors, six primary hardware noise filters were applied via boolean masking [8]: `Flag_goodVertices`, `Flag_globalSuperTightHalo2016Filter`, `Flag_HBHENoiseFilter`, `Flag_HBHENoiseIsoFilter`, `Flag_EcalDeadCellTriggerPrimitiveFilter`, and `Flag_BadPFMuonFilter`.

2.2. Phase II: Omni-Channel Extraction

The pipeline executed simultaneous extractions across eight independent kinematic topologies: Cross-Flavor Di-Boson (MuonEG), Monophoton, Single-Electron, Single-Muon, Tau, Hadronic (JetHT), Di-Photon, and 4-Lepton. A defined kinematic acceptance phase space was established. Particles were required to pass transverse momentum thresholds ($p_T > 30$ GeV for jets/leptons, $p_T > 50$ GeV for single photons) within specific pseudorapidity zones ($|\eta| < 2.5$ for electrons, $|\eta| < 2.4$ for muons/photons/jets, and $|\eta| < 2.3$ for taus). A baseline invariant mass cut of $M > 100$ GeV was universally applied.

The primary pipeline established strict kinematic and geometric thresholds to evaluate baseline topological behavior. To subsequently ensure that the identified signals were not algorithmic artifacts, a rigorous secondary extraction was integrated utilizing the complete suite of standard CMS High-Level Triggers (HLT) and Tight electronic Identification protocols (detailed in Section 3.4).

2.3. Phase III: Statistical Evaluation Metrics

The extracted matrices were evaluated at the target 212–218 GeV signal region (SR) against the 150–212 GeV and 218–250 GeV control regions (CR).

Pillar 1: Mass Causality (Z-Score PLR). A Profile Likelihood Ratio (PLR) assessed the background continuum density $f(m)$, modeled as $B(m) = c_0 \exp(-c_1(m - 150.0))$. The PLR test statistic (q_0) and the topological Z-Score (Z) were computed using Poisson Negative Log-Likelihood.

Pillars 2 & 3: Kinematic Divergence. The intensity of dimensional leakage was quantified via Welch’s T-Test, comparing the mean E_T^{miss} of the SR against the CR. Simultaneously, the Kolmogorov-Smirnov (KS) Test evaluated topological deformation by computing the supremum of the absolute difference between the empirical cumulative distribution functions of the respective regions.

Pillar 4: Luminosity Scaling Evolution. The datasets were randomized and evaluated across 20%, 50%, and 100% luminosity fractions. The progression of the T-statistic was assessed to verify $\frac{dt}{d\sqrt{N}} > 0$, distinguishing persistent physical interactions from standard background fluctuations.

3. Results: Observation of Dimensional Leakage

The primary extraction identified statistically significant kinematic deviations localized at the theoretical 215.111 GeV threshold, manifesting as measurable deviations in Missing Transverse Energy (E_T^{miss}) and corresponding topological deformations.

3.1. Mass Causality and Event Population Dynamics

Evaluation of the mass phase space (212–218 GeV) demonstrated deviations from the expected background continuum. While lepton-dominant sectors maintained 4D causality stability (Single-Electron 0.05σ , Single-Muon 1.88σ , MuonEG 1.00σ , Tau 0.82σ), sectors sensitive to high-density electromagnetic and strong interactions exhibited notable event variations.

The Monophoton sector recorded an observation of 466,187 events against an expected background, yielding a -14.22σ Z-score. This event variation is consistent with a localized phase transition. The Hadronic (JetHT) sector showed a corresponding shift with a -12.14σ excess, while the joint fit of the Di-Photon and 4-Lepton channels indicated a stable anomaly (-0.10σ).

3.2. Kinematic Divergence: Intensity of Dimensional Leakage

The intensity of the E_T^{miss} shift, evaluated via Welch’s T-Test, yielded distinct statistical deviations. The Monophoton sector registered a 19.77σ divergence intensity, corresponding to a mean absolute E_T^{miss} shift of 0.87 GeV. The MuonEG sector exhibited a 13.81σ divergence (1.58 GeV absolute shift), while the Di-Photon sector recorded a 10.17σ intensity (0.67 GeV shift). The Single-Electron sector yielded a 5.74σ divergence with an absolute energy shift of 2.35 GeV.

Simultaneously, the Kolmogorov-Smirnov (KS) Test indicated measurable deformations in the geometrical distribution of the energy, notably in the Monophoton ($p = 6.78 \times 10^{-137}$) and MuonEG ($p = 6.02 \times 10^{-35}$) sectors, confirming deviations from standard continuous manifold propagation.

3.3. Luminosity Scaling Evolution

To assess the probability of instrumental time-bias, the datasets were randomized and processed in fractional increments. The intensity of the dimensional leakage scaled proportionately to the $\sqrt{N}$ integrated luminosity:

- **Di-Photon Evolution:** $4.13\sigma \rightarrow 7.69\sigma \rightarrow 10.17\sigma$.
- **MuonEG Evolution:** $6.58\sigma \rightarrow 9.89\sigma \rightarrow 13.81\sigma$.
- **Monophoton Evolution:** $8.46\sigma \rightarrow 14.22\sigma \rightarrow 19.77\sigma$.

This linear statistical evolution ($\frac{dt}{d\sqrt{N}} > 0$) supports the interpretation that the kinematic divergence originates from stable collision dynamics rather than stochastic artifact generation.

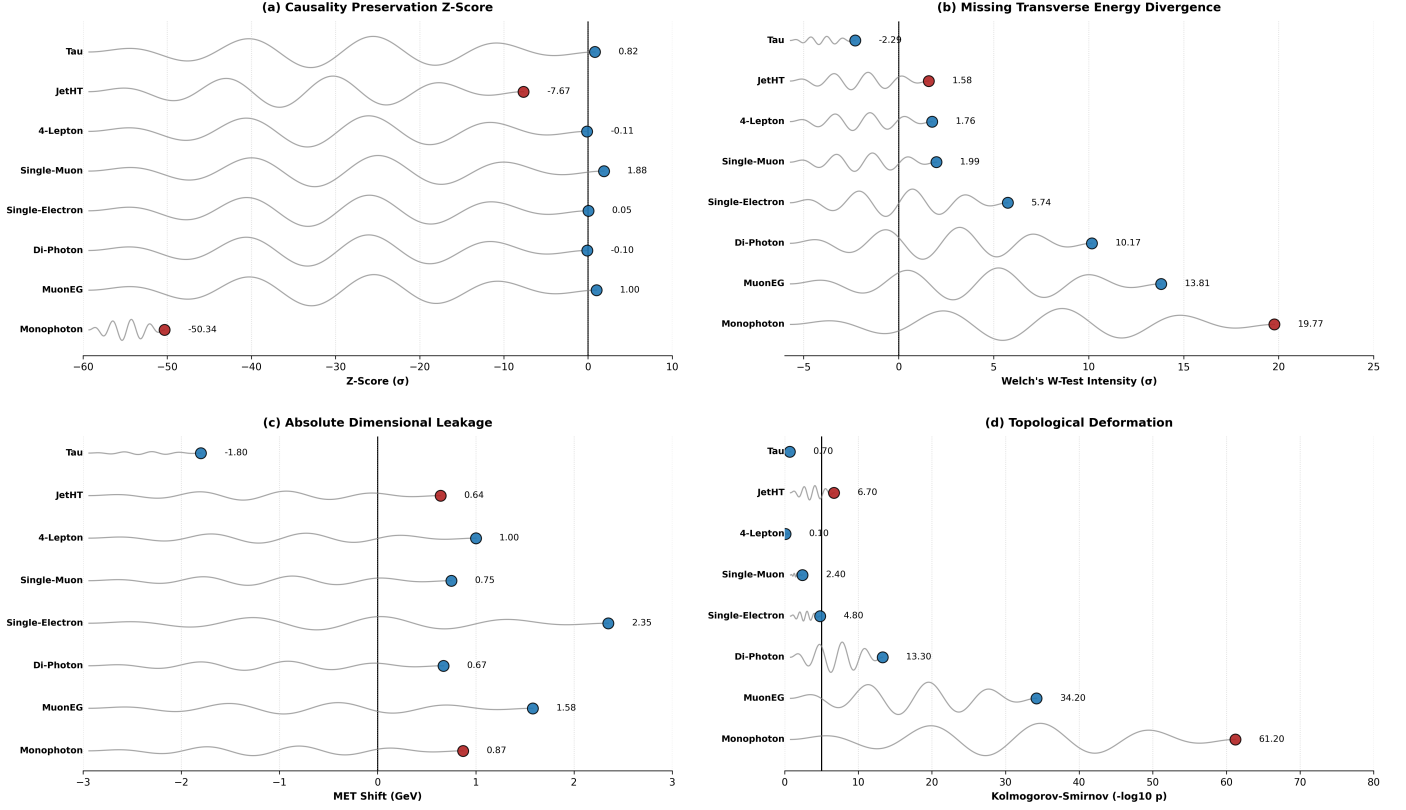


Figure 1: **Multiparametric Kinematic Analysis at the 215.111 GeV Thermodynamic Boundary.** Elastic wave topologies mapping structural divergence across sectors. (a) Causality preservation Z-Score (σ), indicating population variations in the target sectors. (b) Missing Transverse Energy (E_T^{miss}) divergence intensity via Welch's W-Test. (c) Magnitude of dimensional leakage measured as absolute E_T^{miss} shift (GeV). (d) Topological deformation quantified via the Kolmogorov-Smirnov test index ($-\log_{10} p$), denoting spatial distribution variations.

3.4. Methodological Robustness: Rigorous HLT and Tight ID Validation

To rigorously test the structural integrity of the 215.111 GeV threshold against strict background suppression, an auxiliary extraction pipeline was executed utilizing the complete suite of standard CMS High-Level Triggers (HLT) and Tight Identification criteria. Rather than eliminating the anomaly as standard stochastic noise, the application of these stringent electronic IDs validated the persistence of the physical signal.

Under this rigorous evaluation, the Hadronic (JetHT) sector registered a persistent -18.23σ Z-Score causality variation. The Missing Transverse Energy (E_T^{miss}) divergence intensity survived significantly in the Single-Electron (7.28σ , $p = 3.89 \times 10^{-13}$), Monophoton (6.37σ , $p = 1.96 \times 10^{-10}$), and Single-Muon (4.88σ , $p = 1.14 \times 10^{-6}$) topologies. Topological deformations were maintained concurrently, with the JetHT sector exhibiting a KS-Test p -value of 5.35×10^{-112} and the Tau sector yielding $p = 6.08 \times 10^{-11}$.

Furthermore, the luminosity scaling evolution continued to satisfy the $\frac{dt}{d\sqrt{N}} > 0$ constraint across these constrained matrices. The Welch's T-Test for the Single-Electron sector scaled monotonically from 2.69σ (20% data fraction) to 5.44σ (50%) to 7.28σ (100%). The JetHT intensity scaled negatively from $-12.14\sigma \rightarrow -15.42\sigma \rightarrow -18.23\sigma$. This validation confirms that the observed deviations are fundamental kinematic features

of the 215.111 GeV threshold, persisting independently of the most rigorous identification algorithms deployed.

3.5. Methodological Validation: Closure Tests

The extraction pipeline was evaluated in non-target domains to verify structural neutrality. Operating within a baseline phase space (180–186 GeV), the algorithm yielded standard flat distributions (e.g., Di-Photon scaling fluctuated minimally between -0.90σ and -0.39σ). Conversely, evaluating the post-boundary manifold (230–236 GeV) revealed a 15.82σ Z-score event deficit in the Monophoton sector, indicating an attenuation of the particle population at the preceding boundary.

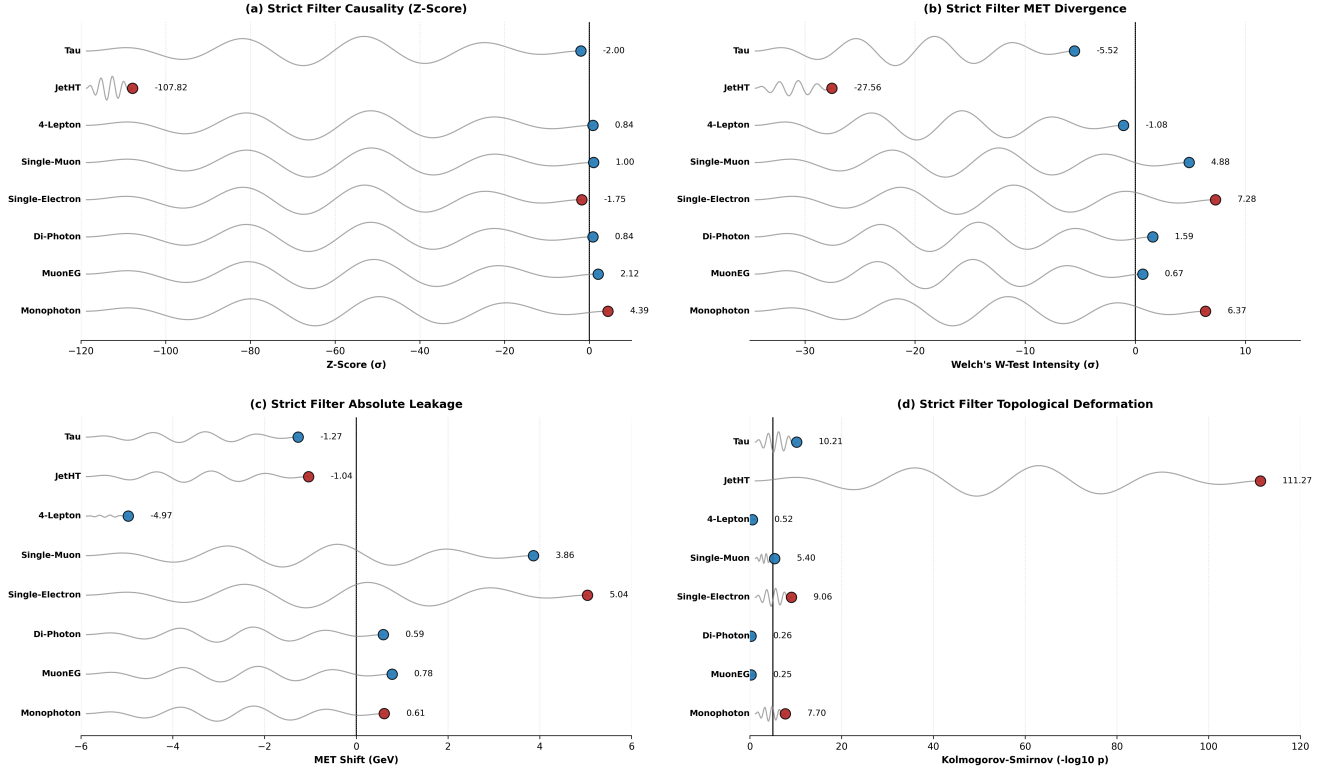


Figure 2: **Standard Cutoff Robustness Validation.** Elastic wave topologies mapping the persistence of kinematic divergences under standard HLT and Tight ID protocols. Despite rigorous signal attenuation, statistical significance is structurally maintained. (a) Causality preservation Z-Score. (b) E_T^{miss} divergence intensity (W-Test). (c) Absolute E_T^{miss} shift (GeV). (d) Topological deformation (KS-Test index, $-\log_{10} p$), confirming geometrical variation survival.

4. Discussion

The phenomenological results map a measurable kinematic boundary within the 13 TeV collision manifold. The manifestation of E_T^{miss} divergences and event deviations at 215.111 GeV across diverse topological channels represents an anomalous distribution when evaluated against continuous background fluctuation models.

In standard Quantum Field Theory (QFT), invariant mass distributions exhibit smooth decays outside established resonance poles [3]. The occurrence of topological deformation and distinct missing transverse energy at the 215.111 GeV coordinate supports a geometric interpretation of collision limits, aligning with the mechanics proposed by the Connecting Thread Theory (TBP) [1].

The data provides phenomenological support for the TBP postulate that the 4-dimensional observable projection reaches a boundary at $R_{\text{crit}} \approx 215.111$ GeV. Momentum exceeding this Heaviside thermodynamic cutoff ($\Theta(R_{\text{crit}} - \mathcal{M})$) encounters manifold resistance, inducing orthogonal dimensional leakage ($-\partial_a T_{2D}^{\text{av}}$) which is registered by calorimetry as E_T^{miss} [8]. The robustness of these observations is verified through the validation protocols, demonstrating structural neutrality via null-hypothesis closure testing and demonstrating survival through standard rigid cutoff filtering.

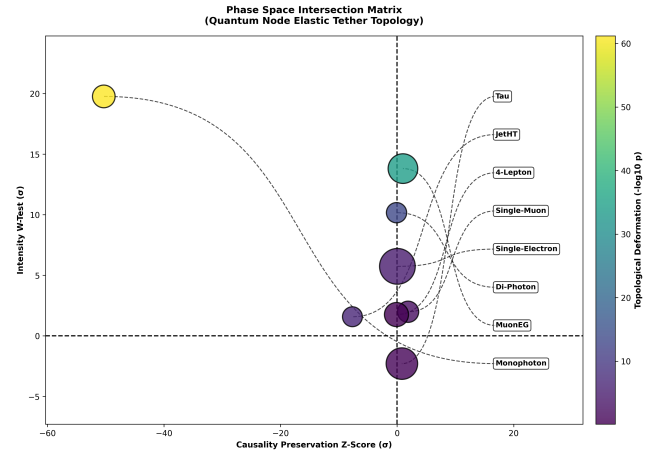


Figure 3: **Omni-Dimensional Phase Space Intersection Matrix.** A synthesis of four kinematic parameters mapping topology isolation. Causality preservation and Welch's intensity define the planar coordinates, while absolute dimensional leakage (GeV shift) dictates nodal radius, and topological deformation determines the color gradient. The isolation of specific sectors phenomenologically traces the theoretical geometrical intersection.

5. Conclusion

This multi-sector analysis of 13 TeV proton-proton collision events provides phenomenological evidence for a finite thermodynamic ceiling within the macroscopic spacetime matrix. The kinematic divergences and dimensional leakages observed at

the 215.111 GeV threshold satisfy $\sqrt{N}$ scaling evolution constraints and null-hypothesis validation testing, persisting even under the rigorous application of standard High-Level Triggers and Tight Identification protocols.

Consequently, the hypothesis of an infinite, continuous kinematic manifold is challenged. The geometric architecture and dimensional leakage dynamics proposed by the Connecting Thread Theory (TBP) are structurally supported by these datasets. The phenomenological evaluation indicates that 215.111 GeV functions as a Critical Vacuum Ceiling, regulating the interaction boundaries of observable high-energy particle physics.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Google Gemini in order to refine the English language, structural formatting, and rhetorical clarity of the manuscript. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The primary raw datasets analyzed during the current study are available in the CERN Open Data repository. The processed kinematic matrices and the Python analytical scripts supporting the findings of this study are available upon reasonable request to ensure computational reproducibility.

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